

Data filtering-based recursive identification for an exponential autoregressive moving average model by using the multi-innovation theory

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Abstract: This study employs the data filtering technique to investigate the recursive identification problems for a non-linear exponential autoregressive model with moving average noise, i.e. the ExpARMA model. Whitening the ExpARMA model by a linear filter, the original identification model is divided into a filtered identification model and a coloured noise model, then a filtering-based extended stochastic gradient algorithm is derived. In order to improve the parameter estimation accuracy, the multi-innovation identification theory is used to develop a filtering-based multi-innovation extended stochastic gradient algorithm for the ExpARMA model. A simulation example is given to demonstrate the superiority of the proposed filtering-based multi-innovation algorithm over the existing algorithms.

1 Introduction

Non-linear random vibrations are common in many observed stochastic processes, such as riverflow discharge, biological populations, economic fluctuations and so on [1]. Although some linear time-series models have been used to approximate these random vibrations, certain non-linear behaviours of random vibrations, including amplitude-dependent frequency, jump phenomena and limit cycles, cannot be well explained by linear models [2]. Therefore, non-linear time-series modelling plays an important role in predicting and controlling random vibrations. The exponential autoregressive (ExpAR) model introduced by Haggan and Ozaki is a typical non-linear time-series model [3, 4]. In the early days, research interest has been focused on the first- or second-order ExpAR model and its stationarity and ergodicity [5, 6]. Later, the general order ExpAR model and some generalised ExpAR models with more difficult structures have attracted much attention. For example, Shi and Aoyama employed the genetic algorithm to estimate the ExpAR model parameters [7]; Ghosh *et al.* transformed the ExpAR model into a non-linear state-space model and used an extended Kalman filter to fit the ExpAR model [8]; Chen *et al.* developed a variable projection method to identify generalised ExpAR models [9]. However, there are few research works on fitting the ExpAR model in the presence of coloured noise, which is generally characterised by autoregressive, moving average (MA) and autoregressive moving average (ARMA) structures [10]. This motivates us to investigate new ExpAR time-series modelling algorithms under the interference of coloured noise.

As a tool of establishing the mathematical models of dynamical systems, system identification has been widely used for non-linear system modelling [11, 12], adaptive control [13], network communication [14] and many other fields. For decades, there have been many contributions on various system identification methods, such as least squares (LS) methods [15, 16], gradient search-based methods [17], maximum likelihood methods [18] and subspace identification methods [19] and so on. Combining with the recursive and iterative techniques, the gradient search-based methods are frequently used for non-linear system identification.

For instance, applying the multi-innovation theory, Meng derived a multi-innovation stochastic gradient algorithm for bilinear systems [20]. With regards to the conventional identification methods for the ExpAR model, the maximum likelihood method is a very time-consuming optimisation procedure and the objective function is not convex thus multiple local optima may exist [8]; the grid-search method is only a procedure by trial and error, without a priori knowledge of the measurement data, a large number of identification experiments have to be carried out [3]. Different from these existing methods, this paper combines the gradient search with some novel identification techniques to explore the identification algorithms for the non-linear ExpAR model with MA noise.

In the field of system identification, the data filtering technique has shown the appropriateness in dealing with coloured noises [21]. Recently, Li and Liu derived a filtering-based maximum likelihood iterative LS algorithm for a class of bilinear systems with ARMA noise [22]. In order to improve the parameter estimation accuracy, the data filtering technique can be combined with other techniques, for example the multi-innovation identification theory [23], to explore novel identification algorithms. In this aspect, Pan *et al.* derived a filtering-based multi-innovation extended stochastic gradient algorithm for multivariable systems with MA noise [24].

Combining the gradient search with the data filtering technique and the multi-innovation identification theory, this paper intends to study recursive identification algorithms for the ExpARMA model. The proposed filtering-based multi-innovation algorithm is based on the work in [25] for the ExpAR model and in [26] for the ExpARMA model, but it extends them in some important ways:

- Inspired by the data filtering technique, we use a linear filter to filter the measurement data and to transform the original identification model into two sub-identification models, that is, the filtered identification model and the noise identification model.
- Combining the gradient search with the data filtering technique, we derive a filtering-based extended stochastic gradient

algorithm for these two sub-identification models whose unknown parameter vectors are estimated interactively.

- Introducing the multi-innovation identification theory, we develop a filtering-based multi-innovation extended stochastic gradient algorithm for the ExpARMA model, such that the parameter estimation accuracy can be improved.

The remainder of this paper is organised as follows. Section 2 describes the identification problem for the ExpARMA model and derives the filtered identification model. Sections 3 and 4 propose the filtering-based extended stochastic gradient (F-ESG) algorithm and the filtering-based multi-innovation extended stochastic gradient (F-MI-ESG) algorithm. For comparison, Section 5 gives the multi-innovation extended stochastic gradient (MI-ESG) algorithm. Section 6 provides a numerical example for testing the effectiveness of the proposed algorithms. Finally, some concluding remarks are given in Section 7.

2 Problem description

Given a time-series $\{y(t), y(t-1), y(t-2), \dots\}$, the n -order ExpAR model with coloured noise can be described as

$$\begin{aligned} y(t) = & [\alpha_1 + \beta_1 e^{-\gamma y^2(t-1)}]y(t-1) \\ & + [\alpha_2 + \beta_2 e^{-\gamma y^2(t-1)}]y(t-2) + \dots \\ & + [\alpha_n + \beta_n e^{-\gamma y^2(t-1)}]y(t-n) + w(t), \end{aligned} \quad (1)$$

where $w(t) := D(z)v(t) \in \mathbb{R}$ is a coloured noise and can be regarded as the output of an MA process driven by a stochastic white noise $v(t)$, $D(z)$ is the polynomial in the unit backward shift operator z^{-1} [$z^{-1}y(t) = y(t-1)$]

$$D(z) := 1 + d_1 z^{-1} + d_2 z^{-2} + \dots + d_m z^{-m}, \quad d_j \in \mathbb{R},$$

and α_i, β_i ($i = 1, 2, \dots, n$), γ and d_j ($j = 1, 2, \dots, m$) are the parameters to be estimated.

The non-linearity of the ExpARMA model comes from the exponential dependence on $\gamma y^2(t-1)$, which makes the dynamics of this process locally linear but globally non-linear. When the parameter γ is large enough, Equation (1) reduces to an ARMA model with respect to the parameters α_i and d_j ; when $\gamma = 0$, (1) reduces to an ARMA model with respect to the parameters $(\alpha_i + \beta_i)$ and d_j . Thus, the parameter γ is introduced to adjust the scaling of the ExpARMA model and should be limited in a certain range, such that $e^{-\gamma y^2(t-1)}$ is different from both zero and one. Moreover, the ExpARMA model can be seen as a smooth threshold model [27, 28], since if $|y(t-1)|$ is large, then (1) is similar to an ARMA model whose parameters are approximately α_i and d_j ; while if $|y(t-1)|$ is small, then the ARMA model parameters switch to $(\alpha_i + \beta_i)$ and d_j .

Assume that the orders m and n are known, the observation data $y(t)$ is measurable. Without loss of generality, the initial values are set to be $y(t) = 0$ and $v(t) = 0$ for $t \leq 0$.

It is obvious that $y(t)$ is linear with respect to the parameters α_i, β_i and d_j , and is non-linear with respect to the parameter γ . Define the parameter vectors

$$\begin{aligned} \theta &:= [\theta^T, d^T]^T \in \mathbb{R}^{2n+m}, \\ \alpha &:= [\alpha^T, \beta^T]^T \in \mathbb{R}^{2n}, \\ \alpha &:= [\alpha_1, \alpha_2, \dots, \alpha_n]^T \in \mathbb{R}^n, \\ \beta &:= [\beta_1, \beta_2, \dots, \beta_n]^T \in \mathbb{R}^n, \\ d &:= [d_1, d_2, \dots, d_m]^T \in \mathbb{R}^m, \end{aligned}$$

and the information vectors:

$$\begin{aligned} \phi(\gamma, t) &:= [\phi_s^T(\gamma, t), \phi_n^T(t)]^T \in \mathbb{R}^{2n+m}, \\ \phi_s(\gamma, t) &:= [\varphi^T(t), e^{-\gamma y^2(t-1)} \varphi^T(t)]^T \in \mathbb{R}^{2n}, \\ \varphi(t) &:= [y(t-1), y(t-2), \dots, y(t-n)]^T \in \mathbb{R}^n, \\ \phi_n(t) &:= [v(t-1), v(t-2), \dots, v(t-m)]^T \in \mathbb{R}^m. \end{aligned}$$

Then, Equation (1) can be written as

$$\begin{aligned} w(t) &= \sum_{j=1}^m d_j v(t-j) + v(t) \\ &= \phi_n^T(t) d + v(t), \end{aligned} \quad (2)$$

$$\begin{aligned} y(t) &= \sum_{i=1}^n \alpha_i y(t-i) + e^{-\gamma y^2(t-1)} \sum_{i=1}^n \beta_i y(t-i) + w(t) \\ &= \varphi^T(t) \alpha + e^{-\gamma y^2(t-1)} \varphi^T(t) \beta + w(t) \\ &= \phi_s^T(\gamma, t) \theta + w(t) \end{aligned} \quad (3)$$

$$\begin{aligned} &= \phi_s^T(\gamma, t) \theta + \phi_n^T(t) d + v(t) \\ &= \phi^T(\gamma, t) \theta + v(t). \end{aligned} \quad (4)$$

Equations (2) and (4) are the noise model and the identification model. The proposed algorithms in this paper are based on this identification model. Many identification methods are derived based on the identification models of the systems [24, 29] and can be used to estimate the parameters of other linear systems and non-linear systems [30–33] and can be applied to fields [34–40] such as chemical process engineering systems. Inspired by the data filtering technique [41], we employ a linear filter to filter the time-series $\{y(t), y(t-1), y(t-2), \dots\}$ and transform the ExpARMA structure into the ExpAR structure. Such a transformation avoids the impact of the coloured noise, and does not change the transfer function but the structure of the original model in (1). Define the linear filter $L(z) := 1/D(z)$ and the filtered variables

$$\begin{aligned} y_f(t) &:= L(z)y(t) = \frac{1}{D(z)}y(t) \\ &= y(t) - [y_f(t-1), y_f(t-2), \dots, y_f(t-m)]d \in \mathbb{R}, \end{aligned} \quad (5)$$

$$\begin{aligned} \varphi_f(t) &:= L(z)\varphi(t) = \frac{1}{D(z)}\varphi(t) \\ &= [y_f(t-1), y_f(t-2), \dots, y_f(t-n)]^T \in \mathbb{R}^n, \end{aligned} \quad (6)$$

$$\begin{aligned} \phi_{sf}(\gamma, t) &:= L(z)\phi_s(\gamma, t) = \frac{1}{D(z)}\phi_s(\gamma, t) \\ &= [\varphi_f^T(t), e^{-\gamma y_f^2(t-1)} \varphi_f^T(t)]^T \in \mathbb{R}^{2n}. \end{aligned} \quad (7)$$

Multiplying both sides of (1) by the linear filter $L(z)$ gives

$$\begin{aligned} y_f(t) &= [\alpha_1 + \beta_1 e^{-\gamma y_f^2(t-1)}]y_f(t-1) \\ &\quad + [\alpha_2 + \beta_2 e^{-\gamma y_f^2(t-1)}]y_f(t-2) + \dots \\ &\quad + [\alpha_n + \beta_n e^{-\gamma y_f^2(t-1)}]y_f(t-n) + v(t) \\ &= \sum_{i=1}^n \alpha_i y_f(t-i) + e^{-\gamma y_f^2(t-1)} \sum_{i=1}^n \beta_i y_f(t-i) + v(t) \\ &= \varphi_f^T(t) \alpha + e^{-\gamma y_f^2(t-1)} \varphi_f^T(t) \beta + v(t) \\ &= \phi_{sf}^T(\gamma, t) \theta + v(t). \end{aligned} \quad (8)$$

This is the filtered identification model and has the classic ExpAR structure. The filtered identification model in (8) contains the parameter sets θ and γ , the noise model in (2) contains the parameter vector d . Thus, all the parameters to be estimated are contained in these two sub-identification models.

Note that the exponential non-linearity exists in both the ExpARMA and ExpAR structures, the identification problems

become difficult and cannot be solved with conventional algorithms, for example, the LS optimisation [42]. The objective of this paper is to study new recursive identification algorithms for the ExpARMA model by the data filtering technique and the multi-innovation identification theory.

3 F-ESG algorithm

In this section, applying the gradient search and the data filtering technique, we derive a F-ESG algorithm for the ExpARMA model.

For the filtered identification model in (8) and the noise model in (2), define two criterion functions

$$J_1(\boldsymbol{\theta}, \gamma) := \frac{1}{2} [y_f(t) - \boldsymbol{\phi}_{sf}^T(\gamma, t) \boldsymbol{\theta}]^2,$$

$$J_2(\mathbf{d}) := \frac{1}{2} [w(t) - \boldsymbol{\phi}_n^T(t) \mathbf{d}]^2.$$

Computing the gradients of $J_1(\boldsymbol{\theta}, \gamma)$ and $J_2(\mathbf{d})$ gives

$$\text{grad}[J_1(\boldsymbol{\theta}, \gamma)] = -\boldsymbol{\psi}_f(t) [y_f(t) - \boldsymbol{\phi}_{sf}^T(\gamma, t) \boldsymbol{\theta}],$$

$$\text{grad}[J_2(\mathbf{d})] = -\boldsymbol{\phi}_n(t) [w(t) - \boldsymbol{\phi}_n^T(t) \mathbf{d}],$$

where the generalised information vector $\boldsymbol{\psi}_f(t)$ is defined as

$$\boldsymbol{\psi}_f(t) := [\boldsymbol{\phi}_{sf}^T(\gamma, t), \boldsymbol{\theta}^T \boldsymbol{\phi}'_{sf}(\gamma, t)]^T \in \mathbb{R}^{2n+1},$$

and the derivative vector $\boldsymbol{\phi}'_{sf}(\gamma, t)$ is defined as

$$\boldsymbol{\phi}'_{sf}(\gamma, t) := \frac{\partial \boldsymbol{\phi}_{sf}(\gamma, t)}{\partial \gamma}$$

$$= [\mathbf{0}_n, -y^2(t-1)e^{-\gamma y^2(t-1)} \boldsymbol{\phi}_f^T(t)]^T \in \mathbb{R}^{2n}.$$

Let $\begin{bmatrix} \hat{\boldsymbol{\theta}}(t) \\ \hat{\gamma}(t) \end{bmatrix} = \begin{bmatrix} \hat{\boldsymbol{\theta}}(t) \\ \hat{\gamma}(t) \end{bmatrix}$ and $\hat{\mathbf{d}}(t)$ represent the estimates of $\begin{bmatrix} \boldsymbol{\theta} \\ \gamma \end{bmatrix}$ and $\mathbf{d}$ at time t . Replacing the unknown parameter sets $\boldsymbol{\theta}$ and γ in $\boldsymbol{\psi}_f(t)$ and $\boldsymbol{\phi}'_{sf}(\gamma, t)$ with their estimates $\hat{\boldsymbol{\theta}}(t-1)$ and $\hat{\gamma}(t-1)$ at the preceding instant, we obtain the estimation vectors

$$\boldsymbol{\psi}_f(t) = [\boldsymbol{\phi}_{sf}^T(\hat{\gamma}(t-1), t), \hat{\boldsymbol{\theta}}^T(t-1) \boldsymbol{\phi}'_{sf}(\hat{\gamma}(t-1), t)]^T, \quad (9)$$

$$\boldsymbol{\phi}'_{sf}(\hat{\gamma}(t-1), t) = [\mathbf{0}_n, -y^2(t-1)e^{-\hat{\gamma}(t-1)y^2(t-1)} \boldsymbol{\phi}_f^T(t)]^T. \quad (10)$$

Minimising $J_1(\boldsymbol{\theta}, \gamma)$ and $J_2(\mathbf{d})$ by the gradient search, we obtain the following recursive relations:

$$\begin{bmatrix} \hat{\boldsymbol{\theta}}(t) \\ \hat{\gamma}(t) \end{bmatrix} = \begin{bmatrix} \hat{\boldsymbol{\theta}}(t-1) \\ \hat{\gamma}(t-1) \end{bmatrix} - \frac{1}{r_1(t)} \text{grad}[J_1(\hat{\boldsymbol{\theta}}(t-1), \hat{\gamma}(t-1))]$$

$$= \begin{bmatrix} \hat{\boldsymbol{\theta}}(t-1) \\ \hat{\gamma}(t-1) \end{bmatrix} + \frac{1}{r_1(t)} \boldsymbol{\psi}_f(t) \Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t-1), \gamma=\hat{\gamma}(t-1)} \quad (11)$$

$$\times [y_f(t) - \boldsymbol{\phi}_{sf}^T(\hat{\gamma}(t-1), t) \hat{\boldsymbol{\theta}}(t-1)],$$

$$r_1(t) = r_1(t-1) + \|\boldsymbol{\psi}_f(t)\|^2_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t-1), \gamma=\hat{\gamma}(t-1)}, \quad (12)$$

$$\hat{\mathbf{d}}(t) = \hat{\mathbf{d}}(t-1) - \frac{1}{r_2(t)} \text{grad}[J_2(\hat{\mathbf{d}}(t-1))]$$

$$= \hat{\mathbf{d}}(t-1) + \frac{1}{r_2(t)} \boldsymbol{\phi}_n(t) [w(t) - \boldsymbol{\phi}_n^T(t) \hat{\mathbf{d}}(t-1)], \quad (13)$$

$$r_2(t) = r_2(t-1) + \|\boldsymbol{\phi}_n(t)\|^2, \quad r_2(0) = 1. \quad (14)$$

However, the recursive computation in (9)–(14) cannot be implemented, since the filtered variables $y_f(t)$, $\boldsymbol{\phi}_{sf}(\hat{\gamma}(t-1), t)$, $\boldsymbol{\psi}_f(t)$ and $\boldsymbol{\phi}'_{sf}(\hat{\gamma}(t-1), t)$ involve the unknown parameters d_j ($j = 1, 2, \dots, m$), and the coloured noise $w(t)$ and the noise

information vector $\boldsymbol{\phi}_n(t)$ are unmeasurable. To address this problem, we construct the estimates for these unknown variables and then replace the unknown variables with their corresponding estimates.

Using the estimate of the noise $v(t-j)$, i.e. $\hat{v}(t-j)$ to construct the estimate of $\boldsymbol{\phi}_n(t)$ gives

$$\hat{\boldsymbol{\phi}}_n(t) := [\hat{v}(t-1), \hat{v}(t-2), \dots, \hat{v}(t-m)]^T \in \mathbb{R}^m.$$

Using $\boldsymbol{\phi}_s(\gamma, t)|_{\gamma=\hat{\gamma}(t-1)}$ and $\hat{\boldsymbol{\phi}}_n(t)$ to define the estimate of $\boldsymbol{\phi}(\gamma, t)|_{\gamma=\hat{\gamma}(t-1)}$, we obtain

$$\hat{\boldsymbol{\phi}}(\hat{\gamma}(t-1), t) := [\boldsymbol{\phi}_s^T(\hat{\gamma}(t-1), t), \hat{\boldsymbol{\phi}}_n^T(t)]^T \in \mathbb{R}^{2n+m}.$$

From (4), we have

$$v(t) = y(t) - \boldsymbol{\phi}^T(\gamma, t) \boldsymbol{\theta}.$$

Replacing $\boldsymbol{\theta}$ and $\boldsymbol{\phi}(\gamma, t)$ with their corresponding estimates $\hat{\boldsymbol{\theta}}(t)$ and $\hat{\boldsymbol{\phi}}(\hat{\gamma}(t-1), t)$ gives

$$\hat{v}(t) = y(t) - \hat{\boldsymbol{\phi}}^T(\hat{\gamma}(t-1), t) \hat{\boldsymbol{\theta}}(t).$$

From (3), we have

$$w(t) = y(t) - \boldsymbol{\phi}_s^T(\gamma, t) \boldsymbol{\theta}.$$

Replacing $\boldsymbol{\theta}$ and γ with their estimates $\hat{\boldsymbol{\theta}}(t-1)$ and $\hat{\gamma}(t-1)$ gives

$$\hat{w}(t) = y(t) - \boldsymbol{\phi}_s^T(\hat{\gamma}(t-1), t) \hat{\boldsymbol{\theta}}(t-1).$$

Using the parameter estimates of the noise identification model, i.e.

$$\hat{\mathbf{d}}(t) := [\hat{d}_1(t), \hat{d}_2(t), \dots, \hat{d}_m(t)]^T \in \mathbb{R}^m$$

to construct the estimate of $D(z)$, we have

$$\hat{D}(t, z) := 1 + \hat{d}_1(t)z^{-1} + \hat{d}_2(t)z^{-2} + \dots + \hat{d}_m(t)z^{-m}.$$

Replacing $D(z)$ in (5)–(7) with $\hat{D}(t, z)$, we obtain the estimates of the filtered variables

$$\hat{y}_f(t) := \frac{1}{\hat{D}(t, z)} y(t)$$

$$= y(t) - [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-m)]^T \hat{\mathbf{d}}(t),$$

$$\hat{\boldsymbol{\phi}}_f(t) := \frac{1}{\hat{D}(t, z)} \boldsymbol{\phi}(t)$$

$$= [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-n)]^T,$$

$$\hat{\boldsymbol{\phi}}_{sf}(\gamma, t) := \frac{1}{\hat{D}(t, z)} \boldsymbol{\phi}_s(\gamma, t)$$

$$= [\hat{\boldsymbol{\phi}}_f^T(t), e^{-\gamma y^2(t-1)} \hat{\boldsymbol{\phi}}_f^T(t)]^T.$$

Replacing $\boldsymbol{\phi}_{sf}(\hat{\gamma}(t-1), t)$ and $\boldsymbol{\phi}_f(t)$ in (9) and (10) with $\hat{\boldsymbol{\phi}}_{sf}(\gamma, t)|_{\gamma=\hat{\gamma}(t-1)}$ and $\hat{\boldsymbol{\phi}}_f(t)$, respectively, we have

$$\hat{\boldsymbol{\psi}}_f(t) = [\hat{\boldsymbol{\phi}}_{sf}^T(\hat{\gamma}(t-1), t), \hat{\boldsymbol{\theta}}^T(t-1) \hat{\boldsymbol{\phi}}_{sf}^T(\hat{\gamma}(t-1), t)]^T,$$

$$\hat{\boldsymbol{\phi}}'_{sf}(\hat{\gamma}(t-1), t) = [\mathbf{0}_n, -y^2(t-1)e^{-\hat{\gamma}(t-1)y^2(t-1)} \hat{\boldsymbol{\phi}}_f^T(t)]^T.$$

Define two innovations

$$e_1(t) := \hat{y}_f(t) - \hat{\boldsymbol{\phi}}_{sf}^T(\hat{\gamma}(t-1), t) \hat{\boldsymbol{\theta}}(t-1) \in \mathbb{R},$$

$$e_2(t) := \hat{w}(t) - \hat{\boldsymbol{\phi}}_n^T(t) \hat{\mathbf{d}}(t-1) \in \mathbb{R}.$$

Replacing the filtered variables $y_f(t)$, $\phi_{sf}(\hat{\gamma}(t-1), t)$, $\psi_f(t)$ and $\phi_{sf}'(\hat{\gamma}(t-1), t)$ in (9)–(12), the coloured noise $w(t)$ and the noise information vector $\phi_n(t)$ in (13) and (14) with their estimates $\hat{y}_f(t)$, $\hat{\phi}_{sf}(\hat{\gamma}(t-1), t)$, $\hat{\psi}_f(t)$, $\hat{\phi}_{sf}'(\hat{\gamma}(t-1), t)$, $\hat{w}(t)$ and $\hat{\phi}_n(t)$, we can summarise the F-ESG algorithm as follows:

$$\begin{bmatrix} \hat{\theta}(t) \\ \hat{\gamma}(t) \end{bmatrix} = \begin{bmatrix} \hat{\theta}(t-1) \\ \hat{\gamma}(t-1) \end{bmatrix} + \frac{1}{r_1(t)} \hat{\psi}_f(t) e_1(t), \quad (15)$$

$$e_1(t) = \hat{y}_f(t) - \hat{\phi}_{sf}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1), \quad (16)$$

$$r_1(t) = r_1(t-1) + \|\hat{\psi}_f(t)\|^2, \quad r_1(0) = 1, \quad (17)$$

$$\hat{d}(t) = \hat{d}(t-1) + \frac{1}{r_2(t)} \hat{\phi}_n(t) e_2(t), \quad (18)$$

$$e_2(t) = \hat{w}(t) - \hat{\phi}_n^T(t) \hat{d}(t-1), \quad (19)$$

$$r_2(t) = r_2(t-1) + \|\hat{\phi}_n(t)\|^2, \quad r_2(0) = 1, \quad (20)$$

$$\hat{\psi}_f(t) = [\hat{\phi}_{sf}^T(\hat{\gamma}(t-1), t), \hat{\theta}^T(t-1) \hat{\phi}_{sf}'(\hat{\gamma}_{t-1}, t)]^T, \quad (21)$$

$$\hat{\phi}_{sf}'(\hat{\gamma}_{t-1}, t) = [\hat{\phi}_f^T(t), e^{-\hat{\gamma}(t-1)y^2(t-1)} \hat{\phi}_f^T(t)]^T, \quad (22)$$

$$\hat{\phi}_{sf}(\hat{\gamma}_{t-1}, t) = [\mathbf{0}_n, -y^2(t-1)e^{-\hat{\gamma}(t-1)y^2(t-1)} \hat{\phi}_f^T(t)]^T, \quad (23)$$

$$\hat{\phi}_f(t) = [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-n)]^T, \quad (24)$$

$$\hat{\phi}_s(\hat{\gamma}_{t-1}, t) = [\hat{\phi}_s^T(\hat{\gamma}(t-1), t), \hat{\phi}_n^T(t)]^T, \quad (25)$$

$$\phi_s(\hat{\gamma}_{t-1}, t) = [\phi^T(t), e^{-\hat{\gamma}(t-1)y^2(t-1)} \phi^T(t)]^T, \quad (26)$$

$$\hat{\phi}_n(t) = [\hat{v}(t-1), \hat{v}(t-2), \dots, \hat{v}(t-m)]^T, \quad (27)$$

$$\phi(t) = [y(t-1), y(t-2), \dots, y(t-n)]^T, \quad (28)$$

$$\hat{y}_f(t) = y(t) - [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-m)]^T \hat{d}(t), \quad (29)$$

$$\hat{w}(t) = y(t) - \hat{\phi}_s^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1), \quad (30)$$

$$\hat{v}(t) = y(t) - \hat{\phi}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t), \quad (31)$$

$$\hat{\theta}(t) = [\hat{\theta}^T(t), \hat{d}^T(t)]^T, \quad (32)$$

$$\hat{d}(t) = [\hat{d}_1(t), \hat{d}_2(t), \dots, \hat{d}_m(t)]^T, \quad (33)$$

$$\hat{\theta}(t) = [\hat{\alpha}_1(t), \dots, \hat{\alpha}_n(t), \hat{\beta}_1(t), \dots, \hat{\beta}_m(t)]^T. \quad (34)$$

The steps of computing the parameter estimates $\hat{\theta}(t)$, $\hat{\gamma}(t)$ and $\hat{d}(t)$ by the F-ESG algorithm in (15)–(34) are listed as follows.

1. Initialisation: let $t = 1$, $\hat{\theta}(0) = \mathbf{1}_{2n}/p_0$, $\hat{\gamma}(0) = 1/p_0$, $\hat{d}(0) = \mathbf{1}_m/p_0$, $\hat{w}(t-l) = 1/p_0$, $\hat{v}(t-l) = 1/p_0$, $\hat{y}_f(t-l) = 1/p_0$, $l = 1, 2, \dots, \max[m, n]$, $p_0 = 10^6$. Give an error tolerance $\varepsilon > 0$.
2. Collect the measurement data $y(t)$, form the information vectors $\phi(t)$, $\hat{\phi}_n(t)$, $\phi_s(\hat{\gamma}(t-1), t)$ and $\hat{\phi}_{sf}'(\hat{\gamma}(t-1), t)$ by (25)–(28).
3. Compute the coloured noise estimate $\hat{w}(t)$ by (30), the reciprocal of the step-size $r_2(t)$ by (20), and the innovation $e_2(t)$ by (19).

4. Update the noise parameter estimation vector $\hat{d}(t)$ by (18). Compute the filtered variable $\hat{y}_f(t)$ by (29).
5. Form the filtered information vectors $\hat{\phi}_f(t)$, $\hat{\phi}_{sf}'(\hat{\gamma}(t-1), t)$, $\hat{\phi}_{sf}(\hat{\gamma}(t-1), t)$ and $\hat{\psi}_f(t)$ by (21)–(24).
6. Compute the reciprocal of the step-size $r_1(t)$ by (17), and the innovation $e_1(t)$ by (16).
7. Update the parameter sets $\hat{\theta}(t)$ and $\hat{\gamma}(t)$ by (15). Construct the parameter estimation vector $\hat{\theta}(t)$ by (32). Compute the noise estimate $\hat{v}(t)$ by (31).
8. If $\|\hat{\theta}(t) - \hat{\theta}(t-1)\| + \|\hat{\gamma}(t) - \hat{\gamma}(t-1)\| + \|\hat{d}(t) - \hat{d}(t-1)\| > \varepsilon$, increase t by 1 and go to step 2; otherwise, output the parameter estimates $\hat{\alpha}_i(t)$ and $\hat{\beta}_i(t)$ from $\hat{\theta}(t)$ by (34), the parameter estimates $\hat{d}_j(t)$ from $\hat{d}(t)$ by (33), and terminate this procedure.

For the F-ESG algorithm, both the innovations $e_1(t)$ and $e_2(t)$ in (16) and (19) are the scalars, which makes the recursive computation only involve the current innovations and measurement data. In order to make full use of the data, we apply the multi-innovation identification theory [43], and derive a filtering-based MI-ESG algorithm for the ExpARMA model in the following section.

4 F-MI-ESG algorithm

Inspired by the multi-innovation identification theory [44], we expand the scalar innovations $e_1(t)$ and $e_2(t)$ into two innovation vectors and derive an F-MI-ESG algorithm for the ExpARMA model. The details are as follows.

Let p denote the innovation length. Define the stacked filtered vector $\hat{Y}_f(p, t)$, the stacked filtered information matrix $\hat{\Phi}_{sf}(\hat{\gamma}(t-1), t)$, the stacked noise vector $\hat{W}(p, t)$ and the stacked noise information matrix $\hat{\Phi}_n(\hat{\gamma}(t-1), t)$ as

$$\begin{aligned} \hat{Y}_f(p, t) &:= [\hat{y}_f(t), \hat{y}_f(t-1), \dots, \hat{y}_f(t-p+1)]^T \in \mathbb{R}^p, \\ \hat{\Phi}_{sf}(\hat{\gamma}_{t-1}, t) &:= [\hat{\phi}_{sf}^T(\hat{\gamma}(t-1), t), \hat{\phi}_{sf}^T(\hat{\gamma}(t-1), t-1), \dots, \\ &\quad \hat{\phi}_{sf}^T(\hat{\gamma}(t-1), t-p+1)] \in \mathbb{R}^{(2m) \times p}, \\ \hat{W}(p, t) &:= [\hat{w}(t), \hat{w}(t-1), \dots, \hat{w}(t-p+1)]^T \in \mathbb{R}^p, \\ \hat{\Phi}_n(p, t) &:= [\hat{\phi}_n(t), \hat{\phi}_n(t-1), \dots, \\ &\quad \hat{\phi}_n(t-p+1)] \in \mathbb{R}^{m \times p}, \end{aligned}$$

and expand the scalar innovations $e_1(t)$ and $e_2(t)$ in (16) and (19) into the innovation vectors

$$E_1(p, t) := \hat{Y}_f(p, t) - \hat{\Phi}_{sf}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1) \in \mathbb{R}^p, \quad (35)$$

$$E_2(p, t) := \hat{W}(p, t) - \hat{\Phi}_n^T(p, t) \hat{d}(t-1) \in \mathbb{R}^p. \quad (36)$$

In order to guarantee the compatibility of the matrix multiplication dimension, the generalised information vector $\hat{\psi}_f(t)$ in (15) must be expanded into a generalised stacked information matrix

$$\hat{\Psi}_f(p, t) := [\hat{\psi}_f(t), \hat{\psi}_f(t-1), \dots, \hat{\psi}_f(t-p+1)] \in \mathbb{R}^{(2n+1) \times p}.$$

Thus, (15) and (18) can be written into the matrix form:

$$\begin{aligned} \begin{bmatrix} \hat{\theta}(t) \\ \hat{\gamma}(t) \end{bmatrix} &= \begin{bmatrix} \hat{\theta}(t-1) \\ \hat{\gamma}(t-1) \end{bmatrix} + \frac{1}{r_1(t)} \hat{\Psi}_f(p, t) E_1(p, t), \\ \hat{d}(t) &= \hat{d}(t-1) + \frac{1}{r_2(t)} \hat{\Phi}_n(p, t) E_2(p, t). \end{aligned}$$

Combining the above equations with (35) and (36), we can summarise the F-MI-ESG algorithm as

$$\begin{bmatrix} \hat{\theta}(t) \\ \hat{\gamma}(t) \end{bmatrix} = \begin{bmatrix} \hat{\theta}(t-1) \\ \hat{\gamma}(t-1) \end{bmatrix} + \frac{1}{r_1(t)} \hat{\Psi}_f(p, t) E_1(p, t), \quad (37)$$

$$E_1(p, t) = \hat{Y}_f(p, t) - \hat{\Phi}_{sf}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1), \quad (38)$$

$$r_1(t) = r_1(t-1) + \|\hat{\Psi}_f(t)\|^2, \quad r_1(0) = 1, \quad (39)$$

$$\hat{d}(t) = \hat{d}(t-1) + \frac{1}{r_2(t)} \hat{\Phi}_n(p, t) E_2(p, t), \quad (40)$$

$$E_2(p, t) = \hat{W}(p, t) - \hat{\Phi}_n^T(p, t) \hat{d}(t-1), \quad (41)$$

$$r_2(t) = r_2(t-1) + \|\hat{\Phi}_n(t)\|^2, \quad r_2(0) = 1, \quad (42)$$

$$\hat{\Psi}_f(p, t) = [\hat{\Psi}_f(t), \hat{\Psi}_f(t-1), \dots, \hat{\Psi}_f(t-p+1)], \quad (43)$$

$$\hat{Y}_f(p, t) = [\hat{y}_f(t), \hat{y}_f(t-1), \dots, \hat{y}_f(t-p+1)]^T, \quad (44)$$

$$\hat{\Phi}_{sf}(\hat{\gamma}_{t-1}, t) = [\hat{\Phi}_{sf}(\hat{\gamma}(t-1), t), \hat{\Phi}_{sf}(\hat{\gamma}(t-1), t-1), \dots, \hat{\Phi}_{sf}(\hat{\gamma}(t-1), t-p+1)], \quad (45)$$

$$\hat{W}(p, t) = [\hat{w}(t), \hat{w}(t-1), \dots, \hat{w}(t-p+1)]^T, \quad (46)$$

$$\hat{\Phi}_n(p, t) = [\hat{\Phi}_n(t), \hat{\Phi}_n(t-1), \dots, \hat{\Phi}_n(t-p+1)], \quad (47)$$

$$\hat{\Psi}_f(t) = [\hat{\Phi}_{sf}^T(\hat{\gamma}(t-1), t), \hat{\theta}^T(t-1) \hat{\Phi}_{sf}'(\hat{\gamma}(t-1), t)]^T, \quad (48)$$

$$\hat{\Phi}_{sf}(\hat{\gamma}_{t-1}, t) = [\hat{\Phi}_f^T(t), e^{-\hat{\gamma}(t-1)y^2(t-1)} \hat{\Phi}_f^T(t)]^T, \quad (49)$$

$$\hat{\Phi}_{sf}(\hat{\gamma}_{t-1}, t) = [\mathbf{0}_n, -y^2(t-1)e^{-\hat{\gamma}(t-1)y^2(t-1)} \hat{\Phi}_f^T(t)]^T, \quad (50)$$

$$\hat{\Phi}_f(t) = [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-n)]^T, \quad (51)$$

$$\hat{\Phi}(\hat{\gamma}_{t-1}, t) = [\hat{\Phi}_s^T(\hat{\gamma}(t-1), t), \hat{\Phi}_n^T(t)]^T, \quad (52)$$

$$\hat{\Phi}_s(\hat{\gamma}_{t-1}, t) = [\varphi^T(t), e^{-\hat{\gamma}(t-1)y^2(t-1)} \varphi^T(t)]^T, \quad (53)$$

$$\hat{\Phi}_n(t) = [\hat{v}(t-1), \hat{v}(t-2), \dots, \hat{v}(t-m)]^T, \quad (54)$$

$$\varphi(t) = [y(t-1), y(t-2), \dots, y(t-n)]^T, \quad (55)$$

$$\hat{y}_f(t) = y(t) - [\hat{y}_f(t-1), \hat{y}_f(t-2), \dots, \hat{y}_f(t-m)]^T \hat{d}(t), \quad (56)$$

$$\hat{w}(t) = y(t) - \hat{\Phi}_s^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1), \quad (57)$$

$$\hat{v}(t) = y(t) - \hat{\Phi}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t), \quad (58)$$

$$\hat{\theta}(t) = [\hat{\theta}^T(t), \hat{d}^T(t)]^T, \quad (59)$$

$$\hat{d}(t) = [\hat{d}_1(t), \hat{d}_2(t), \dots, \hat{d}_m(t)]^T, \quad (60)$$

$$\hat{\theta}(t) = [\hat{\alpha}_1(t), \dots, \hat{\alpha}_n(t), \hat{\beta}_1(t), \dots, \hat{\beta}_n(t)]^T. \quad (61)$$

When $p = 1$, the F-MI-ESG algorithm reduces to the F-ESG algorithm.

The steps of computing the parameter estimates $\hat{\theta}(t)$, $\hat{\gamma}(t)$ and $\hat{d}(t)$ by the F-MI-ESG algorithm in (37)–(61) are listed as follows.

1. Initialisation: Set the innovation length p . Let $t = 1$, $\hat{\theta}(0) = \mathbf{1}_{2n}/p_0$, $\hat{\gamma}(0) = 1/p_0$, $\hat{d}(0) = \mathbf{1}_m/p_0$, $\hat{w}(t-l) = 1/p_0$,

$\hat{v}(t-l) = 1/p_0$, $\hat{y}_f(t-l) = 1/p_0$, $l = 1, 2, \dots, \max[m, n]$, $p_0 = 10^6$. Give an error tolerance $\varepsilon > 0$.

2. Collect the measurement data $y(t)$, form the information vectors $\varphi(t)$, $\hat{\Phi}_n(t)$, $\hat{\Phi}_s(\hat{\gamma}(t-1), t)$ and $\hat{\Phi}(\hat{\gamma}(t-1), t)$ by (52)–(55).
3. Compute the coloured noise estimate $\hat{w}(t)$ by (57). Form the stacked noise vector $\hat{W}(p, t)$ and the stacked noise information matrix $\hat{\Phi}_n(p, t)$ by (46) and (47).
4. Compute the innovation vector $E_2(p, t)$ and the reciprocal of the step-size $r_2(t)$ by (41) and (42).
5. Update the noise parameter estimation vector $\hat{d}(t)$ by (40). Compute the filtered variable $\hat{y}_f(t)$ by (56).
6. Form the filtered information vectors $\hat{\Phi}_f(t)$, $\hat{\Phi}_{sf}(\hat{\gamma}(t-1), t)$, $\hat{\Phi}_{sf}'(\hat{\gamma}(t-1), t)$ and $\hat{\Psi}_f(t)$ by (48)–(51).
7. Construct the generalised stacked information matrix $\hat{\Psi}_f(p, t)$, the stacked filtered vector $\hat{Y}_f(p, t)$ and the stacked filtered information matrix $\hat{\Phi}_{sf}(\hat{\gamma}(t-1), t)$ by (43)–(45).
8. Compute the innovation vector $E_1(p, t)$ and the reciprocal of the step-size $r_1(t)$ by (38) and (39).
9. Update the parameter sets $\hat{\theta}(t)$ and $\hat{\gamma}(t)$ by (37). Construct the parameter estimation vector $\hat{\theta}(t)$ by (59). Compute the noise estimate $\hat{v}(t)$ by (58).
10. If $\|\hat{\theta}(t) - \hat{\theta}(t-1)\| + \|\hat{\gamma}(t) - \hat{\gamma}(t-1)\| + \|\hat{d}(t) - \hat{d}(t-1)\| > \varepsilon$, increase t by 1 and go to step 2; otherwise, output the parameter estimates $\hat{\alpha}_i(t)$ and $\hat{\beta}_i(t)$ from $\hat{\theta}(t)$ by (61), the parameter estimates $\hat{d}_j(t)$ from $\hat{d}(t)$ by (60), and terminate this procedure.

Compared with the F-ESG algorithm in (15)–(34), the F-MI-ESG algorithm involves not only the current data and innovations but also the past $(p-1)$ data and innovations because of the innovation expansion. In addition, $(p-1)$ data and innovations are repeatedly used in the adjacent recursions for the F-MI-ESG algorithm. Thus, the F-MI-ESG algorithm can improve the parameter estimation accuracy.

Remark 1: To initialise the F-MI-ESG algorithm, the initial values $\hat{\theta}(0)$, $\hat{d}(0)$ and $\hat{\gamma}(0)$ are generally arbitrary. For example, $\hat{\theta}(0)$ and $\hat{d}(0)$ can be taken as small real vectors, $\hat{\gamma}(0)$ can be taken as a small number, that is, $\hat{\theta}(0) = \mathbf{1}_{2n}/p_0$, $\hat{\gamma}(0) = 1/p_0$ and $\hat{d}(0) = \mathbf{1}_m/p_0$ with p_0 being a large positive number, e.g. $p_0 = 10^6$. Different p_0 value do not affect the stability performance of the F-MI-ESG algorithm.

Remark 2: Before using the proposed F-MI-ESG algorithm to identify the ExpARMA model, the orders m and n need to be determined from measurement data by using the order estimation methods, such as the orthogonalisation procedure, the correlation analysis and some model order determination criterions in [9].

5 MI-ESG algorithm

As a comparison, this section simply gives a MI-ESG algorithm to show the advantages of the F-MI-ESG algorithm in (37)–(61).

For convenience, we let $\Theta := [\vartheta^T, \gamma^T] \in \mathbb{R}^{2n+m+1}$. Then, the parameter vector Θ involves all the parameters to be estimated. For the identification model in (4), we can obtain the following MI-ESG algorithm by applying the gradient search and multi-innovation theory:

$$\hat{\Theta}(t) = \hat{\Theta}(t-1) + \frac{1}{r(t)} \hat{\Psi}(p, t) E(p, t), \quad (62)$$

$$E(p, t) = Y(p, t) - \hat{\Phi}^T(\hat{\gamma}(t-1), t) \hat{\theta}(t-1), \quad (63)$$

Table 1 MI-ESG estimates and errors ($\sigma^2 = 0.20^2$)

p	t	a_1	a_2	b_1	b_2	γ	d_1	$\delta, \%$
1	100	-0.01432	-0.27281	-0.01433	-0.27281	-0.00073	-0.04017	84.65212
	200	0.02303	-0.28216	0.02303	-0.28215	-0.00169	-0.03771	83.01008
	500	0.03950	-0.31806	0.03950	-0.31807	-0.00284	-0.05403	80.32351
	1000	0.05856	-0.33291	0.05859	-0.33291	-0.00373	-0.06567	78.42273
	2000	0.07250	-0.34254	0.07255	-0.34255	-0.00427	-0.07554	76.98495
	3000	0.07956	-0.34558	0.07963	-0.34559	-0.00448	-0.08269	76.09761
	4000	0.08015	-0.34738	0.08021	-0.34739	-0.00446	-0.09052	75.36477
2	100	-0.01124	-0.30379	-0.01123	-0.30377	-0.00099	-0.09861	78.94126
	200	0.05874	-0.30332	0.05876	-0.30331	0.00018	-0.09635	76.06024
	500	0.07407	-0.34620	0.07409	-0.34618	-0.00222	-0.13589	71.49792
	1000	0.09936	-0.35777	0.09940	-0.35776	-0.00177	-0.16315	68.15780
	2000	0.11516	-0.36387	0.11521	-0.36385	-0.00218	-0.18537	65.69326
	3000	0.12285	-0.36420	0.12291	-0.36418	-0.00236	-0.20060	64.12689
	4000	0.12007	-0.36451	0.12012	-0.36449	-0.00299	-0.21595	62.83523
6	100	0.02020	-0.31876	0.02032	-0.31868	0.00298	-0.28491	61.43541
	200	0.14967	-0.31681	0.14971	-0.31666	0.00241	-0.29604	55.43144
	500	0.12993	-0.35234	0.13002	-0.35217	0.00012	-0.38702	47.46935
	1000	0.16289	-0.37477	0.16300	-0.37460	-0.00119	-0.44661	41.29921
	2000	0.18207	-0.37843	0.18218	-0.37826	-0.00248	-0.48804	37.39722
	3000	0.18838	-0.37553	0.18849	-0.37536	-0.00313	-0.51732	34.83649
	4000	0.17733	-0.37507	0.17742	-0.37489	-0.00371	-0.54488	32.74013
true values		0.20000	-0.36000	0.23000	-0.40000	0.15000	-0.85000	

$$r(t) = r(t-1) + \|\hat{\psi}(t)\|^2, \quad r(0) = 1, \quad (64)$$

$$Y(p, t) = [y(t), y(t-1), \dots, y(t-p+1)]^T, \quad (65)$$

$$\hat{\Phi}(\hat{\gamma}_{t-1}, t) = [\hat{\phi}(\hat{\gamma}(t-1), t), \hat{\phi}(\hat{\gamma}(t-1), t-1), \dots, \hat{\phi}(\hat{\gamma}(t-1), t-p+1)], \quad (66)$$

$$\hat{\Psi}(p, t) = [\hat{\psi}(t), \hat{\psi}(t-1), \dots, \hat{\psi}(t-p+1)], \quad (67)$$

$$\hat{\psi}(t) = [\hat{\phi}^T(\hat{\gamma}(t-1), t), \hat{\theta}^T(t-1)\hat{\phi}^T(\hat{\gamma}(t-1), t)]^T, \quad (68)$$

$$\hat{\phi}(\hat{\gamma}_{t-1}, t) = [\hat{\phi}_s^T(\hat{\gamma}(t-1), t), \hat{\phi}_n^T(t)]^T, \quad (69)$$

$$\hat{\phi}_s(\hat{\gamma}_{t-1}, t) = [\varphi^T(t), e^{-\hat{\gamma}(t-1)y^2(t-1)}\varphi^T(t)]^T, \quad (70)$$

$$\varphi(t) = [y(t-1), y(t-2), \dots, y(t-n)]^T, \quad (71)$$

$$\hat{\phi}_n(t) = [\hat{v}(t-1), \hat{v}(t-2), \dots, \hat{v}(t-m)]^T, \quad (72)$$

$$\hat{\phi}'(\hat{\gamma}_{t-1}, t) = [\mathbf{0}_n, -y^2(t-1)e^{-\hat{\gamma}(t-1)y^2(t-1)}\varphi^T(t), \mathbf{0}_m]^T, \quad (73)$$

$$\hat{v}(t) = y(t) - \hat{\phi}^T(\hat{\gamma}(t-1), t)\hat{\theta}(t), \quad (74)$$

$$\hat{\Theta}(t) = [\hat{\theta}^T(t), \hat{\gamma}(t)]^T, \quad (75)$$

$$\hat{\theta}(t) = [\hat{\alpha}_1(t), \dots, \hat{\alpha}_n(t), \hat{\beta}_1(t), \dots, \hat{\beta}_n(t), \hat{d}_1(t), \dots, \hat{d}_m(t)]^T. \quad (76)$$

Compared with the MI-ESG algorithm in (62)–(76), the proposed F-MI-ESG algorithm in (37)–(61) estimates the filtered identification model parameters and the MA noise model parameters by an interactive way and avoids the influence of the MA noise. The proposed approaches in the paper can combine other mathematical tools and statistical strategies to study the performances of some parameter estimation algorithms of other linear stochastic systems and non-linear stochastic systems with different structures and disturbance noises [45–53] and can be applied to literatures [54–61] such as paper-making systems.

6 Example

Consider the following ExpARMA model:

$$\begin{aligned} y(t) &= [\alpha_1 + \beta_1 e^{-\gamma y^2(t-1)}]y(t-1) \\ &\quad + \dots + [\alpha_n + \beta_n e^{-\gamma y^2(t-n)}]y(t-n) + D(z)v(t) \\ &= [0.20 + 0.23e^{-0.15y^2(t-1)}]y(t-1) \\ &\quad + [-0.36 - 0.40e^{-0.15y^2(t-1)}]y(t-2) \\ &\quad - 0.85v(t-1) + v(t). \end{aligned}$$

The parameters to be estimated are

$$\begin{aligned} \Theta &= [\alpha_1, \alpha_2, \beta_1, \beta_2, d_1, \gamma]^T \\ &= [0.20, -0.36, 0.23, -0.40, -0.85, 0.15]^T. \end{aligned}$$

In simulation, the noise $\{v(t)\}$ is taken as a white noise sequence with zero mean and variance σ^2 , and the data length is taken as $L_e = 4000$.

To exhibit the advantages of the proposed filtering-based multi-innovation identification algorithm, we fix the noise variance $\sigma^2 = 0.20^2$ and use the MI-ESG algorithm and the F-MI-ESG algorithm with $p = 1$, $p = 2$ and $p = 6$ to identify this ExpARMA model, respectively. The parameter estimates and their errors are shown in Tables 1 and 2, the parameter estimation errors $\delta := \|\hat{\Theta}(t) - \Theta\| / \|\Theta\| \times 100\%$ against t are shown in Fig. 1. The F-MI-ESG estimates against t are shown in Figs. 2 and 3.

To show the effect of the noise level on the proposed F-MI-ESG algorithm, we fix the innovation length $p = 6$ and use the F-MI-ESG algorithm with $\sigma^2 = 0.20^2$, $\sigma^2 = 1.50^2$ and $\sigma^2 = 2.30^2$ to identify this ExpARMA model, respectively. The parameter estimation errors δ against t are shown in Fig. 4.

From Tables 1 and 2 and Figs. 1–4, we can draw the following conclusions:

1. The parameter estimation errors given by the MI-ESG algorithm and the F-MI-ESG algorithm become smaller as t increases – see Tables 1, 2 and Fig. 1.
2. For the same innovation length p , the proposed F-MI-ESG algorithm has higher parameter estimation accuracy than the MI-ESG algorithm – see Tables 1, 2 and Fig. 1.

Table 2 F-MI-ESG estimates and errors ($\sigma^2 = 0.20^2$)

p	t	a_1	a_2	b_1	b_2	γ	d_1	$\delta, \%$
1	100	0.00188	-0.26636	0.00186	-0.26635	-0.00102	-0.15278	74.56314
	200	0.04434	-0.28386	0.04435	-0.28386	-0.00266	-0.13016	74.12239
	500	0.06224	-0.32795	0.06225	-0.32797	-0.00420	-0.22296	64.58369
	1000	0.08579	-0.34676	0.08586	-0.34679	-0.00617	-0.27648	58.82362
	2000	0.10418	-0.35807	0.10430	-0.35811	-0.00753	-0.31360	54.85544
	3000	0.11390	-0.36235	0.11405	-0.36240	-0.00829	-0.34849	51.48275
	4000	0.11671	-0.36480	0.11687	-0.36485	-0.00848	-0.39123	47.70376
2	100	0.02465	-0.30595	0.02512	-0.30573	0.04421	-0.30194	59.27077
	200	0.09807	-0.31986	0.09793	-0.31923	0.05069	-0.28283	57.09212
	500	0.11721	-0.36612	0.11716	-0.36521	0.06764	-0.44002	41.66190
	1000	0.14770	-0.38162	0.14657	-0.38027	0.07540	-0.52592	32.77929
	2000	0.16935	-0.38735	0.16764	-0.38583	0.07981	-0.57463	27.70315
	3000	0.18030	-0.38878	0.17826	-0.38713	0.08158	-0.62399	23.03420
	4000	0.18079	-0.38859	0.17873	-0.38700	0.08171	-0.68521	17.81956
6	100	0.06419	-0.31597	0.06720	-0.31598	0.07734	-0.60550	32.48794
	200	0.17796	-0.35568	0.18074	-0.35302	0.11706	-0.67325	18.22806
	500	0.18830	-0.35891	0.19324	-0.35775	0.12315	-0.69751	15.54544
	1000	0.20394	-0.39164	0.20721	-0.38819	0.14595	-0.78244	7.44918
	2000	0.21768	-0.39850	0.21996	-0.39489	0.15482	-0.79673	6.51324
	3000	0.22785	-0.39359	0.22951	-0.38949	0.15475	-0.83335	4.53080
	4000	0.21894	-0.38829	0.22095	-0.38491	0.14860	-0.88826	5.10123
true values		0.20000	-0.36000	0.23000	-0.40000	0.15000	-0.85000	

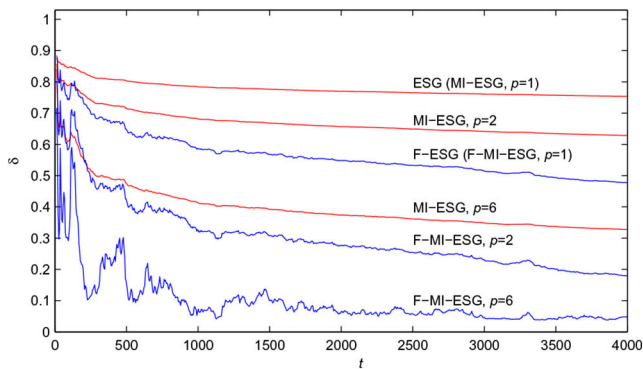


Fig. 1 MI-ESG and F-MI-ESG estimation errors δ against t ($\sigma^2 = 0.20^2$)

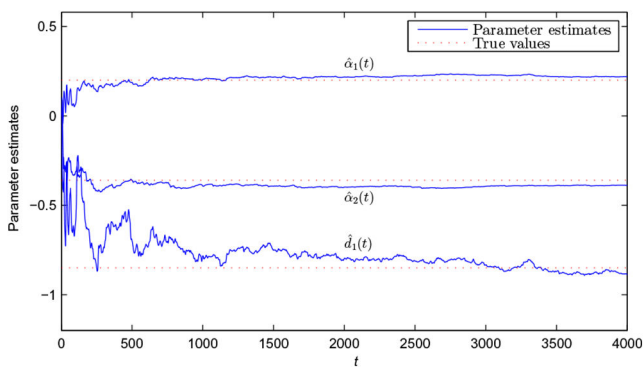


Fig. 2 F-MI-ESG estimates $\hat{\alpha}_1(t)$, $\hat{\alpha}_2(t)$ and $\hat{d}_1(t)$ against t ($\sigma^2 = 0.20^2$, $p = 6$)

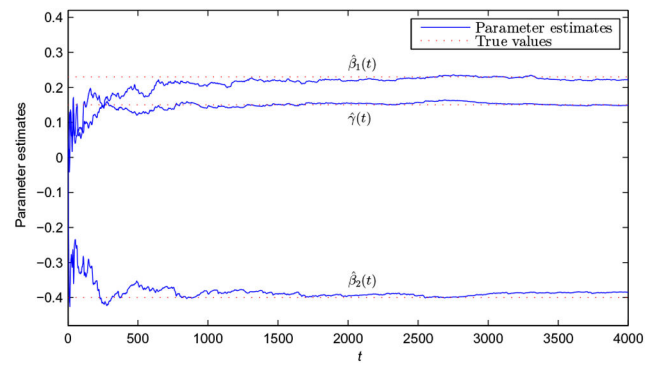


Fig. 3 F-MI-ESG estimates $\hat{\beta}_1(t)$, $\hat{\beta}_2(t)$ and $\hat{\gamma}(t)$ against t ($\sigma^2 = 0.20^2$, $p = 6$)

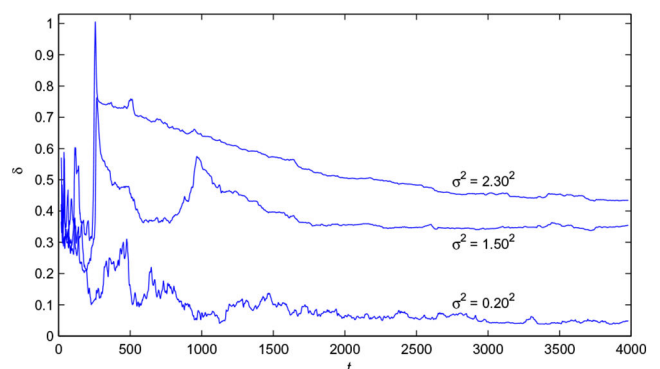


Fig. 4 F-MI-ESG estimation errors δ against t ($p = 6$)

- Compared with the F-EGS algorithm, the F-MI-ESG algorithm has improved estimation accuracy and the estimation accuracy becomes higher with p increasing – see Table 2 and Fig. 1.
- With an appropriate innovation length, the proposed F-MI-ESG parameter estimates approach their corresponding true values and the F-MI-ESG algorithm can identify the ExpARMA model – see Figs. 2 and 3.

- The estimation errors of the proposed F-MI-ESG algorithm become smaller with the decreasing of the noise levels – see Fig. 4.

To compare the performance of the F-MI-ESG algorithm proposed in this paper and the existing LS algorithm in [3], we use the remaining $L_r = 200$ data from $t = L_e + 1$ to $t = L_e + L_r$ and the estimated models obtained by the F-MI-ESG algorithm with $p = 6$ and by the LS algorithm for the model validation. The F-MI-ESG

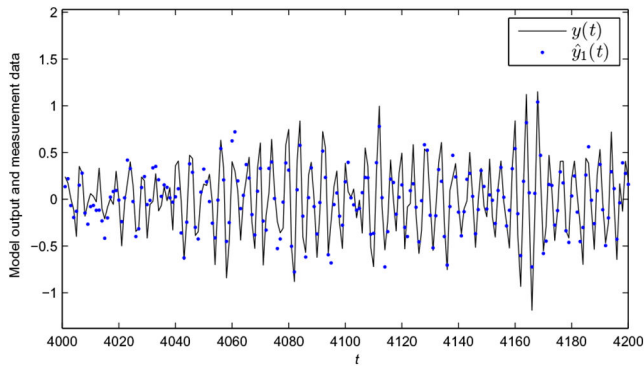


Fig. 5 Model output $\hat{y}_1(t)$ and measurement data $y(t)$

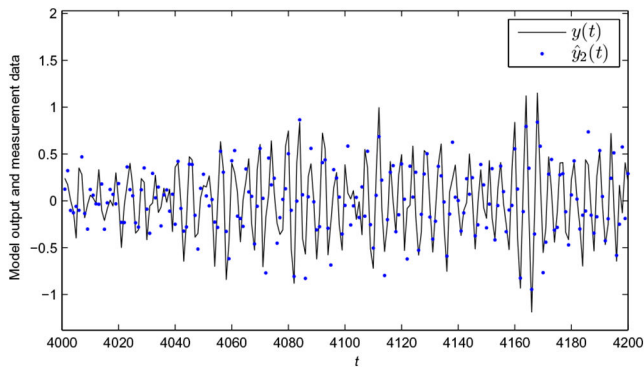


Fig. 6 Model output $\hat{y}_2(t)$ and measurement data $y(t)$

model output $\hat{y}_1(t)$, LS model output $\hat{y}_2(t)$ and the measurement data $y(t)$ are plotted in Figs. 5 and 6. To evaluate the performance, we define and compute the mean square errors (MSEs)

$$\text{MSE}_1 := \left[\frac{1}{L_r} \sum_{t=L_e+1}^{L_e+L_r} [\hat{y}_1(t) - y(t)]^2 \right]^{1/2} = 0.20886,$$

$$\text{MSE}_2 := \left[\frac{1}{L_r} \sum_{t=L_e+1}^{L_e+L_r} [\hat{y}_2(t) - y(t)]^2 \right]^{1/2} = 0.28675.$$

From Figs. 5 and 6, we can see that the model output $\hat{y}_1(t)$ is closer to the measurement data $y(t)$, which means that the estimated model by the F-MI-ESG algorithm can capture the dynamics of this ExpARMA model.

Remark 3: From the simulation results, we can see that the estimation errors of both the F-ESG algorithm and the F-MI-ESG algorithm become smaller as the data t increases. Compared with the F-ESG algorithm, the F-MI-ESG algorithm uses more measurement data and innovations at each recursion. Thus, the estimation errors of the F-MI-ESG algorithm are smaller than that of the F-ESG algorithm and close to zero at the end of the recursion.

7 Conclusions

For the ExpARMA model, the F-ESG algorithm is derived by using the data filtering technique and the gradient search. Applying the multi-innovation identification theory, we expand the scalar innovations into the innovation vectors and develop the F-MI-ESG algorithm. The simulation results demonstrate that introducing the data filtering technique and the multi-innovation identification theory can improve the parameter estimation accuracy. The proposed filtering-based recursive identification algorithm in this paper can be applied to other control and schedule areas such as the information processing and transportation communication systems [62–68] and so on.

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